Darwin to write: ‘I cannot persuade myself that a beneficent and omnipotent God would have designedly created the Ich-neumonidae with the express intention of their feeding within the living bodies of Caterpillars … ’ He might as well have used the example of a French chef boiling lobsters alive to preserve the flavour. Returning to the life of the female digger wasp, it is a solitary one except that other females are working independently in the same area, and sometimes they occupy one another’s burrows rather than go to the trouble of digging a new one.

Dr Jane Brockmann is a sort of wasp equivalent of Jane Goodall. She came from America to work with me at Oxford, bringing her copious records of almost every event in the life of two entire populations of individually identified female wasps. These records were so complete that individual wasp time-budgets could be drawn up. Time is an economic commodity: the more time spent on one part of life, the less is available for other parts. Alan Grafen joined the two of us and taught us how to think correctly about time-costs and reproductive benefits. We found evidence for a true mixed ESS in a game played between female wasps in a population in New Hampshire, though we failed to find such evidence in another population in Michigan. Briefly, the New Hampshire wasps either dig their own nests or enter a nest that another wasp has dug. According to our interpretation, wasps can benefit by entering because some nests are abandoned by their original diggers and are reusable. It does not pay to enter a nest that is occupied, but an enterer has no way of knowing which nests are occupied and which abandoned. She runs the risk of going for days in double-occupation, at the end of which she may come home to find the burrow sealed up, and all her efforts in vain—the other occupant has laid her egg and will reap the benefits. If too much entering is going on in a population, available burrows become scarce, the chance of double-occupation goes up, and it therefore pays to dig. Conversely, if plenty of wasps are digging, the high availability of burrows favours entering. There is a critical frequency of entering in the population at which digging and entering are equally profitable. If the actual frequency is below the critical frequency, natural selection favours entering, because there is a good supply of available abandoned burrows. If the actual frequency is higher than the critical frequency, there is a shortage of available burrows and natural selection favours digging. So a balance is maintained in the population. The detailed, quantitative evidence suggests that this is a true mixed ESS, each individual wasp having a probability of digging or entering, rather than the population containing a mixture of digging and entering specialists.

p. 104 *The neatest demonstration I know of this form of behavioural asymmetry …*

An even clearer demonstration than Tinbergen’s of the ‘resident always wins’ phenomenon comes from N. B. Davies’s research on speckled wood butterflies. Tinbergen’s work was done before ESS theory was invented, and my ESS interpretation in the first edition of this book was made with hindsight. Davies conceived his butterfly study in the light of ESS theory. He noticed that individual male butterflies in Wytham Wood, near Oxford, defended patches of sunlight. Females were attracted to sun patches, so a sun patch was a valuable resource, something worth fighting over. There were more males than sun patches and the surplus waited their chance in the leafy canopy. By catching males and releasing them one after the other, Davies showed that whichever of two individuals was released first into a sun patch was treated, by both individuals, as the ‘owner’. Whichever male arrived second in the sun patch was treated as the ‘intruder’. The intruder always, without exception, promptly conceded defeat, leaving the owner in sole control. In a final *coup de grâce* experiment, Davies managed to ‘fool’ both butterflies into ‘thinking’ that they were the owner and the other was the intruder. Only under these conditions did a really serious, prolonged fight break out. By the way, in all those cases where, for simplicity, I have spoken as though there was a single pair of butterflies there was really, of course, a statistical sample of pairs.

p. 106 *Paradoxical ESS*